

Generation Interconnection System Impact Study Report

For Queue Project Q20-1609

Greenland Gap 345 kV

March 05, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1609. The Interconnection Customer proposes a 437.3 MW solar photovoltaic facility interconnecting at the Greenland Gap 345 kV substation in the West Virginia area. The request was received on November 16, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 541 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,657,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Artesia Country Club 345/1kV Autotransformer	Air Force Base – Artesia Country Club 345kV	115.2	680	5.0
Artesia Country Club 138-1 kV XFMR	Air Force Base – Artesia Country Club 345kV L	115.2	680	5.0
International Paper Valliant Ok – Kearneyville 345kV Line	St. John – Rio DII 138kV	110.5	1,675	12.1
Prides Corner – Auburn Rd 345kV L	Firstenegy Fort Martin Power Station – Ronco 345kV	101.9	1,662	8.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Artesia Country Club 345/1kV Autotransformer	Replace transformer with higher-rated unit	\$25,933,000	26% (\$6,677,000)
Artesia Country Club 138-1 kV XFMR	Replace transformer with higher-rated unit	\$4,785,000	26% (\$1,232,000)
International Paper Valliant Ok – Kearneyville 345kV L	Rebuild line with higher-capacity conductor	\$64,347,000	10% (\$6,607,000)
Prides Corner – Auburn Rd 345kV L	Reconductor line	\$4,647,000	3% (\$141,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$14,657,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.